

Qatar-1b

R-1775

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_151002 · created 2026-08-02T22:35:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457297.70235 +/- 0.00065 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.147 +/- 0.024
Transit depth (Rp/Rs)^2	2.17 +/- 0.69 [%]
Orbital Inclination (inc)	83.51 +/- 1.89
Airmass coefficient 1 (a1)	1.189 +/- 0.018
Airmass coefficient 2 (a2)	-0.135 +/- 0.026
Scatter in the residuals of the lightcurve fit is	1.49 %
Best Comparison Star	None
Optimal Aperture	2.83
Optimal Annulus	10.51
Transit Duration (day)	0.062 +/- 0.015

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.147 ± 0.024 vs 0.1463 (deviation 0.03σ)
Duration fit vs published	0.062 d vs 0.0692 d (deviation 0.43σ)
Mid-time O–C	1.3 min
Depth SNR	5.5
Residual scatter	1.49 %

Light curve

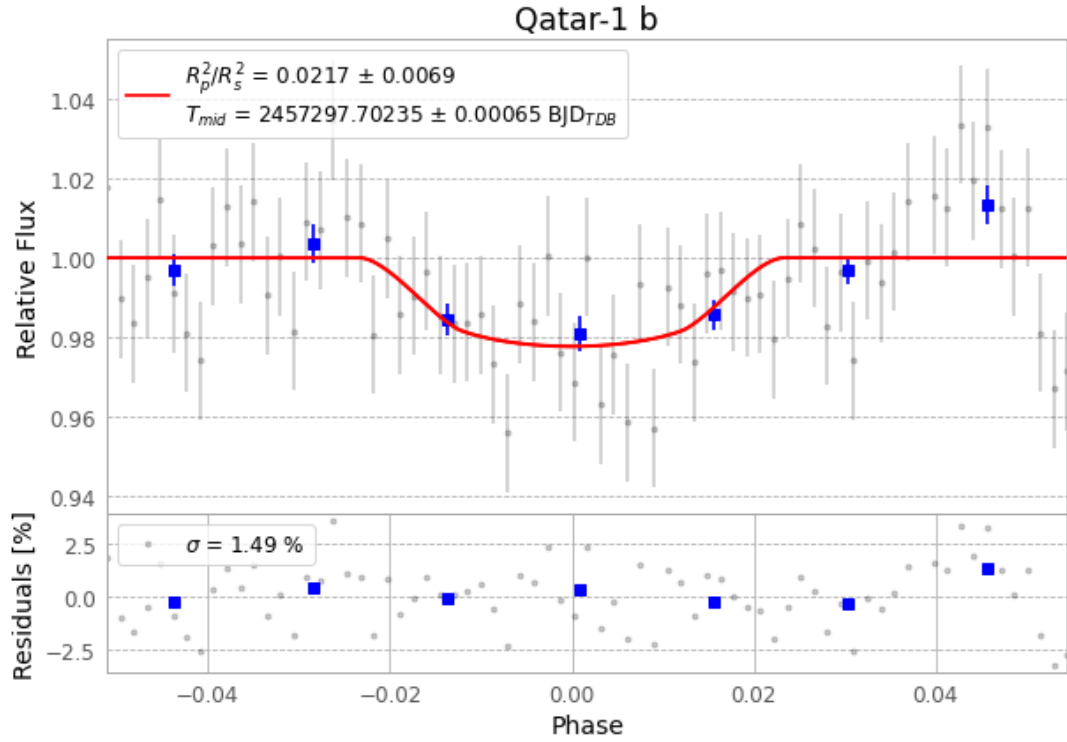


Figure 1 — FinalLightCurve_Qatar-1 b_2015-10-01.png (SHA-256 93ec95d422acaa72...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [355, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dc8fadeea32d18b6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), Qatar-1151002030316.FITS → Qatar-1151002063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-151002.log (8d203df47d12...), FinalLightCurve_Qatar-1 b_2015-10-01.png (93ec95d422ac...), FinalLightCurve_Qatar-1 b_2015-10-01.csv (dac1e2b484f9...)

Compute resources

Wall time 175.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 22:35Z.